



KIASORT Installation Guideline and Requirements

Download KIASORT from:

<https://github.com/banaiek/KIASORT>

Add the downloaded folder to your MATLAB path

In MATLAB:

Run: `kiaSort.m`

MATLAB Requirements for KIASORT

System Requirements

- **MATLAB 2021b or later** (required for optimal performance)
- **Operating Systems:** Windows 10/11, macOS 10.15+, Ubuntu 20.04+

Required MATLAB Toolboxes

- Signal Processing Toolbox
- Statistics and Machine Learning Toolbox
- Parallel Computing Toolbox (recommended for performance)
- Image Processing Toolbox
- Curve Fitting Toolbox
- Deep Learning Toolbox

Python Requirements for KIASORT

Python Installation

- **Python 3.8 - 3.11** (recommended: Python 3.10)
- **Required packages:**
 - `numpy` $\geq 1.20.0$
 - `umap-learn` $\geq 0.5.0$
 - `scikit-learn` $\geq 1.0.0$
 - `numba` $\geq 0.56.0$

Quick Installation

If not installed download Python from python.org

Install packages:

```
pip install numpy umap-learn scikit-learn numba
```

MATLAB-Python Configuration

1. Find Python path:

- Windows: `where python`
- macOS/Linux: `which python3`

2. Configure in MATLAB:

```
% Set Python environment  
pyenv('Version', '/path/to/python')  
  
% Verify setup  
pe = pyenv;  
disp(pe.Version)
```

Quick Verification Test

Run in MATLAB to verify everything works:

```
% Test Python integration

py.sys.version

% Test required modules

numpy = py.importlib.import_module('numpy');
umap_module = py.importlib.import_module('umap.umap_');
```

Common Issues & Solutions

Python Issues

Problem	Solution
"Python not found"	Add Python to PATH, restart MATLAB
"Module not found"	Run <code>pip install umap-learn numpy</code>
Version conflicts	Use Python 3.8-3.11, avoid 3.12+
Permission errors (macOS)	Use <code>sudo pip install</code> or homebrew

Matlab Issues

Problem	Solution
<code>pyenv</code> fails	Use full Python path in <code>pyenv('Version', 'path')</code>
Import errors	Restart MATLAB after Python setup

Memory issues	Ensure 16GB+ RAM for large datasets
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Troubleshooting Checklist

Before running KIASORT, verify:

- MATLAB 2021b+ with required toolboxes
- Python 3.8-3.11 installed and in PATH
- MATLAB can access Python (pyenv shows "Loaded: true")
- Sufficient system memory (16GB+ recommended)

System Recommendations

Recommended Configuration

- 16GB+ RAM
- 8-core CPU
- SSD storage
- GPU